higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# NATIONAL CERTIFICATE

# POWER MACHINES N6

(8190046)

## 2 December 2019 (X-Paper)
## 09:00–12:00

**REQUIREMENTS: Properties of water and steam (BOE 173)**
**Superheated steam tables (appendix to BOE 173)**

**Nonprogrammable calculators and drawing instruments may be used.**

**This question paper consists of 7 pages and a formula sheet of 5 pages.**

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# DEPARTMENT OF HIGHER EDUCATION AND TRAINING
# REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
POWER MACHINES N6
TIME: 3 HOURS
MARKS: 100

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**INSTRUCTIONS AND INFORMATION**

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Questions may be answered in any order, but keep subsections of questions together.
5. ALL sketches and diagrams must be neat, fully labelled and drawn in pencil in the ANSWER BOOK.
6. Write down ALL formulae used.
7. Show ALL intermediate steps for calculations.
8. Use only a black or blue pen.
9. Approximate final answers correctly to THREE decimals unless stated otherwise.
10. Write neatly and legibly.

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## QUESTION 1

The following data was recorded during a test on an open-circuit, continuous-combustion, constant-pressure gas-turbine plant:

| | |
|---|---|
| Air temperature at compressor inlet | 21 °C |
| Air temperature at compressor outlet | 256 °C |
| Air temperature at combustion chamber inlet | 412 °C |
| Temperature of combustion products at turbine inlet | 789 °C |
| Temperature of combustion products at turbine outlet | 502 °C |
| Temperature at plant exit | 221 °C |
| Turbine pressure ratio | 5:1 |
| Mass flow rate of air | 7 kg/s |
| Specific heat capacity of air | 1,005 kJ/kg.K |
| Gamma for air | 1,4 |

Ignore the mass of fuel used and calculate:

1.1 The power required to compress the air in kW (3)

1.2 The power developed by the turbine in kW (3)

1.3 The energy received from the combustion chamber in kW (3)

1.4 The thermal efficiency of the plant (2)

1.5 The efficiency of the heat exchanger (3)

1.6 The absolute isentropic temperature of the turbine (3)

1.7 The isentropic efficiency of the turbine (3)

**[20]**

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## QUESTION 2

A three-stage, single-acting, reciprocating compressor delivers 16 kg of air per minute to an after-cooler whilst rotating at 300 r/min. The index for expansion and compression is 1,32 and intercooling is complete for maximum efficiency. The initial conditions are a pressure of 105 kPa at a temperature of 21 °C. The diameter of the intermediate cylinder is 240 mm with a stroke length of 296 mm. The geometric stage pressure ratio is 4:1.

Assume R for air as 0,287 kJ/kg.K and calculate:

2.1 The volume of air drawn into the low-pressure cylinder in $m^3$/cycle (3)

2.2 For the intermediate cylinder:

2.2.1 The volume of air drawn in, in $m^3$/cycle, correct to FOUR decimal places (3)

2.2.2 The swept volume in $m^3$/cycle correct to FOUR decimal places (3)

2.2.3 The volumetric efficiency (2)

2.2.4 The clearance volume in $m^3$/cycle correct to FIVE decimal places (3)

2.2.5 The cylinder volume in $m^3$/cycle correct to FIVE decimal places (2)

2.3 The power required to drive the compressor in kW (4)

**[20]**

## QUESTION 3

One kilogram of air at a pressure of 110 kPa is compressed isentropically to a pressure of 490 kPa with a temperature of 198 °C. Heat is then transferred to the air at constant pressure until the temperature is 420 °C greater than the initial cycle temperature. The air is then expanded isentropically to the initial pressure. Finally, heat is transferred at constant pressure from the air, until initial conditions are reached.

Assume gamma for air as 1,4, the $C_p$ for air as 1,005 kJ/kg.K and calculate:

| | | |
|---|---|---|
| 3.1 | The initial absolute air temperature | (3) |
| 3.2 | The absolute air temperature after constant-pressure heat addition | (2) |
| 3.3 | The heat received by the air in kJ | (3) |
| 3.4 | The absolute air temperature after isentropic expansion | (3) |
| 3.5 | The heat rejected by the air in kJ | (3) |
| 3.6 | The value of $C_v$ | (2) |
| 3.7 | The value of R | (2) |
| 3.8 | The thermal efficiency of the cycle | (2) |

**[20]**

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## QUESTION 4

A single-cylinder, single-acting compressor of a vapour-compression refrigeration plant, operating between temperature limits of -6 °C and 27 °C, has a swept volume of 1 500 cm³. The volumetric efficiency of the compressor is 90%, while the mechanical efficiency is 80%. The power required to drive the compressor is 9,25 kW, whilst rotating at 122 r/min. At the compressor inlet, the ammonia is 90% dry, while at the compressor outlet, it is superheated. The specific heat capacity for superheated ammonia is 1,814 kJ/kg.K.

The following are extracts from ammonia tables:

| Saturation temperature [°C] | Specific volume [m³/kg] | | Specific enthalpy [kJ/kg] | |
|---|---|---|---|---|
| | Liquid [$v_f$] | Vapour [$v_g$] | Liquid [$h_f$] | Vapour [$h_g$] |
| -6 | 0,00105 | 0,0122 | 72 | 320,5 |
| 27 | 0,00140 | 0,00413 | 159,7 | 279,9 |

Use the given data to calculate:

4.1 The effective swept volume of the compressor in m³ (2)

4.2 The mass flow rate of the ammonia in kg/min (3)

4.3 The work done by the compressor in kJ/kg (3)

4.4 The specific enthalpy of the ammonia at the compressor inlet (3)

4.5 The specific enthalpy of the ammonia at the compressor outlet (2)

4.6 The temperature of the ammonia after compression in °C (4)

4.7 The heat extracted from the condenser in kJ/min (3)

**[20]**

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## QUESTION 5

NO axial thrust was developed in the first stage and second stage of a velocity-compounded, two-stage impulse turbine. The inlet angle of the fixed blades is 35°, while the average blade velocity is 152 m/s. The steam exits the moving blades of the second stage at an angle of 30° to the plane of rotation of the blades. The velocity of flow at the inlet to the second stage is 90 m/s and the velocity coefficient for all the blades is 0,96.

5.1 Construct, to a scale of 1 mm = 4 m/s, velocity diagrams for the turbine in the ANSWER BOOK.

Indicate the lengths of ALL lines as well as the magnitude of ALL angles on the diagrams. (10)

5.2 Determine from the velocity diagrams drawn in QUESTION 5.1:

5.2.1 The velocity of the steam entering the fixed blades in m/s (1)

5.2.2 The velocity of the steam leaving the fixed blades in m/s (1)

5.2.3 The inlet angle of the first row of moving blades (1)

5.2.4 The exit angle of the first row of moving blades (1)

5.2.5 The inlet angle of the second row of moving blades (1)

5.2.6 The nozzle angle (1)

5.2.7 The outlet angle of the fixed blades (1)

5.2.8 The power developed in the turbine in kJ/kg steam (3)

**[20]**

**TOTAL: 100**

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## FORMULA SHEET

This formulae sheet may NOT necessarily be complete. Any formulae used by candidates, which do NOT appear on this list, must be written in full in the ANSWER BOOK.

ENGLISH GENERAL AFRIKAANS

$$P_a V_a = mRT_a$$

$$R = C_p - C_v$$

$$\gamma = \frac{C_p}{C_v}$$

$PV = c$ $PV = k$

$PV^n = c$ $PV^n = k$

$PV^\gamma = c$ $PV^\gamma = k$

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{(n-1)} = \left(\frac{P_2}{P_1}\right)^{\left(\frac{n-1}{n}\right)}$$

$$\Delta U = m.C_v.\Delta T$$

$Q = \Delta U + Wd$ $Q = \Delta U + Av$

$$\Delta s = m\left[C_v \,.\, \ln\left(\frac{P_2}{P_1}\right) + C_p \,.\, \ln\left(\frac{V_2}{V_1}\right)\right]$$

$$\Delta s = m \,.\, C_v \,.\, \ln\left(\frac{P_2}{P_1}\right)$$

$$\Delta s = m.C_p.\ln\left(\frac{V_2}{V_1}\right)$$

$$\Delta s = m \,.\, R \,.\, \left(\frac{P_1}{P_2}\right)$$

$$Q = m \,.\, C_p \,.\, \Delta T$$

$$Q = m \,.\, C_v \,.\, \Delta T$$

$$S_{su} = S_g + C_p.\ln\left(\frac{T_{su}}{T_s}\right)$$

$$S_{fg} = S_g - S_f$$

$$S = S_f + x \,.\, S_{fg}$$

$$h_{su} = h_g + C_p \,.\, \left(t_{su} - t_s\right)$$

$$T_a\left(s_a - s_b\right) = h_a - h_b$$

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ENGLISH

$$h_{ws} = h_f + x.h_{fg}$$

$$V_{ws} = x.V_g$$

GENERAL

$$V_{su} = \frac{\frac{n-1}{n}\left(h_{su} - 1\,941\right)}{P_{su}}$$

$$r = \frac{V_s + V_c}{V_c}$$

$$V_s = \frac{\pi}{4} \times d^2 \times L$$

$$P_2 = \sqrt{P_1 \times P_3}$$

$$r_{ps} = \sqrt[x]{\frac{P_{x+1}}{P_1}}$$

AFRIKAANS

$$h_{ns} = h_f + x.h_{fg}$$

$$V_{ns} = x.V_g$$

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Different formulae for work done (Wd)

verskillende formules vir arbeid verrig (Av)

$$= P \times \Delta V$$

$$= P_1 V_1 . \ln\left(\frac{V_2}{V_1}\right)$$

$$= \frac{P_1 V_1 - P_2 V_2}{n-1}$$

$$= \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$$

$$= m.C_p \,.\, \Delta T$$

$$= \frac{xn}{n-1} \times P_1 V_e \left[\left(\frac{P_{x+1}}{P_1}\right)^{\left(\frac{n-1}{xn}\right)} - 1\right]$$

$$= \frac{xn}{n-1} \times mRtT_1 \left[\left(r_{ps}\right)^{\left(\frac{n-1}{n}\right)} - 1\right]$$

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ENGLISH GENERAL AFRIKAANS

Different formulae for work done (Wd)

Verskillende formules vir arbeid verrig (Av)

= area of PV - diagram

= area van PV - diagram

= work done first stage + work done second stage + ...

= arbeid verrig eerste stadium + arbeid verrig tweede stadium +...

$$Wd_{nett} = Wd_t - Wd_c$$
$$Wd_{nett} = Q_{nett}$$

$$Av_{nett} = Av_t - Av_k$$
$$Av_{nett} = Q_{nett}$$

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Different formulae for air standard efficiencies (ASE)

Verskillende formules vir lugstandaardrendemente (LSR)

$$= 1 - \left(\frac{1}{r}\right)^{(\gamma-1)}$$

$$= 1 - \frac{r_p \cdot (r_c)^{(\gamma-1)}}{r_v^{(\gamma-1)}\left[(r_p - 1) + \gamma^{r_p}(r_c - 1)\right]}$$

$$= \frac{heat\ added - heat\ \ rejected}{heat\ added} \qquad = 1 - \frac{\beta^{\gamma} - 1}{r^{(\gamma-1)} \times \gamma(\beta - 1)} \qquad = \frac{warmte\ toegevoeg - warmte\ afgestaan}{warmte\ toegevoeg}$$

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Different volumetric efficiencies, $\eta_{vol}$

Verskilllende volumetriese rendemente, $\eta_{vol}$

$$= \frac{Volume\ of\ \ air\ taken\ in}{Swept\ volume}$$

$$= \frac{Volume\ lug\ \ ingeneem}{Slagvolume}$$

$$= \frac{Volume\ of\ \ free\ air}{Swept\ volume}$$

$$= \frac{Volume\ vrylug}{Slagvolume}$$

$$= 1 - \left(\frac{V_c}{V_s}\right)\left[\left(\frac{P_2}{P_1}\right)^{\left(\frac{1}{n}\right)} - 1\right]$$

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| ENGLISH | GENERAL | AFRIKAANS |
|---|---|---|
| Different thermal efficiencies, $\eta_{therm.}$ $= \frac{\text{Wd}}{\text{heat supplied}}$ | | Verskillende termiese rendemente, $\eta_{therm.}$ $= \frac{\text{Av}}{\text{warmte toegevoeg}}$ |
| $\eta_{brake\ therm.} = \frac{BP}{m_{f/s} \times CV}$ | | $\eta_{rem\ term.} = \frac{RD}{m_{b/s} \times WW}$ |
| $\eta_{ind.\ therm.} = \frac{IP}{m_{f/s} \times CV}$ | | $\eta_{ind.\ term.} = \frac{ID}{m_{b/s} \times WW}$ |
| $\eta_{therm.} = \frac{m_s(h_s - h_w)}{m_f \times CV}$ | | $\eta_{term.} = \frac{m_s(h_s - h_w)}{m_b \times WW}$ |
| $\eta_c = \frac{T_2' - T_1}{T_2 - T_1}$ | $\eta_t = \frac{T_3 - T_4}{T_3 - T_4'}$ | $\eta_k = \frac{T_2' - T_1}{T_2 - T_1}$ |
| $\eta_{mech.} = \frac{BP}{IP}$ | | $\eta_{meg.} = \frac{RD}{ID}$ |
| Indicated efficiency ratio $= \frac{\eta_{ind.\ therm.}}{ASE}$ | | Indikateurrendementverhouding $= \frac{\eta_{ind.\ term.}}{LSR}$ |
| Brake efficiency ratio $= \frac{\eta_{brake\ therm.}}{ASE}$ | | Remrendementverhouding $= \frac{\eta_{rem.\ term.}}{LSR}$ |
| $BP = 2\pi \frac{TN}{60}$ | $T = F \times r$ | $RD = 2\pi \frac{TN}{60}$ |
| $BP = P_{brake\ mean}\ L \times A \times N \times E$ | | $RD = P_{rem\ gem.}\ L \times A \times N \times E$ |
| $IP = P_{ind.\ mean}\ L \times A \times N \times E$ | | $ID = P_{ind.\ gem.}\ L \times A \times N \times E$ |
| $ISFC = \frac{M_{f/h}}{IP}$ | | $ISBV = \frac{M_{b/h}}{ID}$ |
| $BSFC = \frac{M_{f/h}}{BP}$ | | $RSBV = \frac{M_{b/h}}{RD}$ |
| $COP = \frac{T_1}{T_2 - T_1}$ | | $KVW = \frac{T_1}{T_2 - T_1}$ |
| $COP = \frac{RE}{Wd}$ | | $KVW = \frac{VE}{Av}$ |
| $P = m\ .\ U\ .\ \Delta Vw$ | | $D = m\ .\ U\ .\ \Delta Vw$ |
| $F_{ax.} = m\ .\ \Delta V_f$ | | $F_{aks.} = m\ .\ \Delta V_f$ |

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$$\eta_{dia.} = \frac{2 \,.\, U \,.\, \Delta V_w}{(V_1)^2}$$

$$P_c = P_1\left(\frac{2}{\gamma+1}\right)^{\left(\frac{\gamma}{\gamma-1}\right)}$$

$$T_c = T_1\left(\frac{2}{\gamma+1}\right)$$

$$C_c = \sqrt{2\times10^3\left(h_1 - h_c\right) + (C_1)^2}$$

$$C_2 = \sqrt{2\times10^3\left(h_1 - h_2\right) + (C_1)^2}$$

$$C_c = \sqrt{2\times10^3\times C_p\left(T_1 - T_c\right) + (C_1)^2}$$

$$C_2 = \sqrt{2\times10^3\times C_p\left(T_1 - T_2\right) + (C_1)^2}$$

$$A_c = \frac{m \,.\, V_c}{C_c} \qquad A_2 = \frac{m \,.\, V_2}{C_2}$$

$$\eta = \frac{h_1 - h_c}{h_1 - h_c'} \qquad \eta = \frac{T_1 - T_c}{T_1 - T_c'}$$

$$\eta = \frac{h_c - h_2}{h_c - h_2'} \qquad \eta = \frac{T_c - T_2}{T_c - T_2'}$$

$$\eta = \frac{h_1 - h_2}{h_1 - h_2'} \qquad \eta = \frac{T_1 - T_2}{T_1 - T_2'}$$

$$EE = \frac{m_s\left(h_s - h_w\right)}{m_f \times 2\ 257} \qquad EV = \frac{m_s\left(h_s - h_w\right)}{m_b \times 2\ 257}$$

$$\eta_{iso.} = \frac{Wd_{iso.}}{Wd_{poly.}} \qquad \eta_{iso.} = \frac{Av_{iso.}}{Av_{poli.}}$$

$$\eta_{rank.} = \frac{Wd}{Q} \qquad \eta_{rank.} = \frac{Av}{Q}$$

$$\eta_{carn.} = 1 - \frac{T_2}{T_1}$$

$$h = u + pV$$

$$gZ_1 + U_1 + P_1V_1 + \frac{(C_1)^2}{2} + Q = \qquad gZ_1 + U_1 + P_1V_1 + \frac{(C_1)^2}{2} + Q =$$

$$gZ_2 + U_2 + P_2V_2 + \frac{(C_2)^2}{2} + Wd \qquad gZ_2 + U_2 + P_2V_2 + \frac{(C_2)^2}{2} + Av$$

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